

By substituting this form of wavefunction into the Schrödinger's equation (13.1.1a), we find that the spatially varying part of the wavefunction $u_n(\mathbf{r})$ must satisfy the eigenvalue equation, known as time-independent Schrödinger's equation,

$$\hat{H}_0 u_n(\mathbf{r}) = E_n u_n(\mathbf{r}), \quad (13.1.3)$$

Where $E_n = \hbar\omega_n$, $u_n(\mathbf{r})$ are called the energy eigensolutions of Eq. (13.1.3) and are assumed to be orthonormal in that they obey the relation

$$\int u_m^*(\mathbf{r}) u_n(\mathbf{r}) d^3\mathbf{r} = \delta_{nm}, \quad (13.1.4)$$

The wavefunction of state s , $\psi_s(\mathbf{r}, t)$, can be expressed in terms of eigenstates $u_n(\mathbf{r})$

$$\psi_s(\mathbf{r}, t) = \sum_n C_n^s(t) u_n(\mathbf{r}), \quad (13.1.5)$$

The expansion coefficient gives the probability that the atom is in the energy eigenstate n at time t .



Introducing Eq. (13.1.5) into Schrödinger's equation (13.1.1a), we have

$$i\hbar \sum_n \frac{dC_n^s(t)}{dt} u_n(\mathbf{r}) = \sum_n C_n^s(t) \hat{H} u_n(\mathbf{r}), \quad (13.1.6)$$

Multiplying each side from the left by $u_m^*(\mathbf{r})$, integrate over all space, and utilize Eq. (13.1.4), we have

$$i\hbar \frac{dC_m^s(t)}{dt} = \sum_n H_{mn} C_n^s(t), \quad (13.1.7)$$

Where

$$H_{mn} = \int u_m^*(\mathbf{r}) \hat{H} u_n(\mathbf{r}) d^3\mathbf{r} \quad (13.1.8)$$

The expectation value of any observable quantity can be calculated in terms of wavefunction of the system.

$$\langle A \rangle = \int \Psi_s^* \hat{A} \Psi_s d^3\mathbf{r} \quad (13.1.9)$$



CHAPTER 13

Optical Absorption and Recombination of Bulk Semiconductors

- Introduction to quantum theory
- Optical absorption of semiconductors
- Exciton-enhancement of optical absorption
- Optical recombination



13.1 Introduction to quantum theory

If a quantum-mechanical system, such as an atom, is known to be in a particular quantum-mechanical state, which we designate s , we can describe all of the physical properties of the system in terms of the wavefunction $\psi_s(\mathbf{r}, t)$, which obeys Schrödinger's equation

$$i\hbar \frac{\partial \psi_s(\mathbf{r}, t)}{\partial t} = \hat{H} \psi_s(\mathbf{r}, t), \quad (13.1.1a)$$

$$\hat{H} = \hat{H}_0 + \hat{H}', \quad (13.1.1b)$$

Where $\hat{H}_0$ is the Hamiltonian of a free atom and $\hat{H}'$ is the interaction with the external field. For the case in which no external field is applied to the atom, the Hamiltonian is simply equal to $\hat{H}_0$ and Schrödinger equation possesses solution in the form of energy eigenstates. These states are also known as stationary states, because the time evolution of these states is given by a simple exponential phase factor as follows:

$$\psi_n(\mathbf{r}, t) = u_n(\mathbf{r}) \exp(-i\omega t) \quad (13.1.2)$$



13.2 Optical absorption of semiconductors

13.2.1 Optical absorption of semiconductors – overview

In general, there are several optical absorption processes, and each of these contributes to the total absorption. These processes include:

- Phonon absorption
- Plasmon absorption
- Fundamental absorption
- Exciton absorption
- Absorption due to dopants and imperfections
- Absorption due to intraband transitions
- Free carrier absorption



9

Spectral regions of optical absorption

Table 9.1. Spectral ranges with relevance to semiconductor optical properties

Range	wavelengths	energies
deep ultraviolet	< 250 nm	> 5 eV
ultraviolet	250–400 nm	3–5 eV
visible	400–800 nm	1.6–3 eV
near infrared	800 nm–2 μm	0.6–1.6 eV
mid-infrared	2–20 μm	60 meV–0.6 eV
far infrared	20–80 μm	1.6–60 meV
THz region	> 80 μm	< 1.6 meV

$$E = \frac{1240 \text{ eV nm}}{\lambda}$$



6

This relationship can be conveniently written in Dirac notation

$$\langle A \rangle = \langle \psi_s | \hat{A} | \psi_s \rangle = \langle s | \hat{A} | s \rangle. \quad (13.1.10)$$

The expectation value $\langle A \rangle$ can also be expressed in terms of the probability amplitude. By introducing Eq. (13.1.5) into Eq. (13.1.9), we obtain

$$\langle A \rangle = \sum_{mn} C_m^s C_n^s A_{mn}, \quad (13.1.11)$$

Where

$$A_{mn} = \langle u_m | \hat{A} | u_n \rangle = \int u_m^* \hat{A} u_n d^3 r. \quad (13.1.12)$$



7

Fermi golden rule

Assume the Hamiltonian of the system is H_0 , and is at initial state $|i\rangle$, with a time-dependent perturbation H' , the atom will transit to state $|f\rangle$. The transition probability per time is given by

$$w_{fi}(\hbar\omega) = \frac{2\pi}{\hbar} |H'_{fi}|^2 \delta(E_f - E_i - \hbar\omega), \quad (13.1.13a)$$

where $\hbar\omega$ is the photon energy, E_i (E_f) is the energy of the initial (final) state.

$$H'_{fi} = \langle f | H' | i \rangle \quad (13.1.13b)$$



Optical absorptions of semiconductors

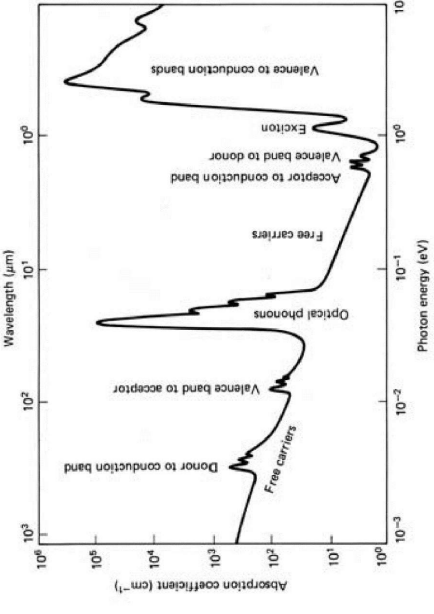
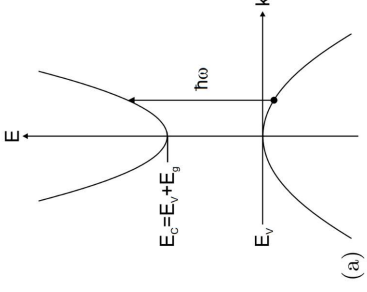


Fig. 9.1.1. Schematic absorption spectrum of a typical semiconductor. With permission from [278], ©1989 Prentice-Hall



$$(13.2.1)$$

$$(13.2.2)$$

(a) Direct transition

- In the fundamental absorption process, a photon of energy greater than the bandgap energy E_g excites an electron from the valence band to the conduction band.
- The minimum of the conduction band and the maximum of the valence band occur at the same value of the wave vector k .
- Both energy and momentum must be conserved in this process. Since the photon momentum is small, the absorption process should essentially conserve the electron momentum

$$\frac{(\hbar \mathbf{k}_i)^2}{2m_h} + \hbar\omega = E_g + \frac{(\hbar \mathbf{k}_f)^2}{2m_e}$$

$$\mathbf{k}_i \approx \mathbf{k}_f$$

Optical absorption of direct bandgap semiconductors

$$R(\hbar\omega) = \frac{2\pi}{\hbar} \int_{\mathbf{k}_c} \int_{\mathbf{k}_v} |\langle c | \mathbf{H}_{em} | v \rangle|^2 \delta(E_c(\mathbf{k}_c) - E_v(\mathbf{k}_v) - \hbar\omega) d\mathbf{k}_c d\mathbf{k}_v \quad (13.2.3a)$$

$$\mathbf{H}_{em} = -\frac{q}{m} \mathbf{A} \cdot \mathbf{p} = \frac{iq\hbar}{m} \mathbf{A} \cdot \nabla \approx q\mathbf{r} \cdot \mathbf{E}.$$

$$|\langle c | \mathbf{H}_{em} | v \rangle|^2 = \frac{e^2 |A|^2}{m^2} |\langle c | \hat{\mathbf{e}} \cdot \mathbf{p} | v \rangle|^2 \quad (13.2.3b)$$

$$\langle c | \hat{\mathbf{e}} \cdot \mathbf{p} | v \rangle^2 = \frac{1}{3} |\mathbf{p}_{cv}|^2 = M_b^2 \quad (13.2.4a)$$

$$R(\hbar\omega) = \frac{2\pi}{\hbar} \left(\frac{e}{m\omega} \right)^2 \frac{|\mathcal{E}(\omega)|^2}{2} |\mathbf{p}_{cv}|^2 \int_{\mathbf{k}} \delta(E_c(\mathbf{k}) - E_v(\mathbf{k}) - \hbar\omega) d\mathbf{k} \quad (13.2.4b)$$

$$\epsilon_1 = \frac{1}{4\pi\epsilon_0} \left(\frac{2\pi e}{m\omega} \right)^2 |\mathbf{p}_{cv}|^2 \int_{\mathbf{k}} \delta(E_c(\mathbf{k}) - E_v(\mathbf{k}) - \hbar\omega) d\mathbf{k} \quad (13.2.5)$$

$$\epsilon_r = 1 + \int_{\mathbf{k}} \frac{e^2}{\epsilon_0 m \omega_{cv}^2} \frac{2|\mathbf{p}_{cv}|^2}{m \hbar \omega_{cv}} \frac{1}{1 - \omega^2/\omega_{cv}^2} d\mathbf{k}, \quad (13.2.6a)$$

$$f = \frac{e^2}{\epsilon_0 m \omega_{cv}^2} \frac{2|\mathbf{p}_{cv}|^2}{m \hbar \omega_{cv}} \quad (13.2.6b)$$

Joint Density of States (JDOS): it quantifies the number of electron-hole pairs available for interband optical transitions between the valence band (VB) and conduction band (CB) of a semiconductor.

13.2.2 Fundamental absorption of semiconductors - Band-band absorption

- When the minimum of the conduction band and the maximum of the valence band occur at the same value of the wave vector k , transitions are direct, and the semiconductor is referred to as a direct-gap semiconductor.
- If the band extrema do not occur at the same wave vector k , transitions are indirect, and the semiconductor is referred to as an indirect-gap semiconductor.

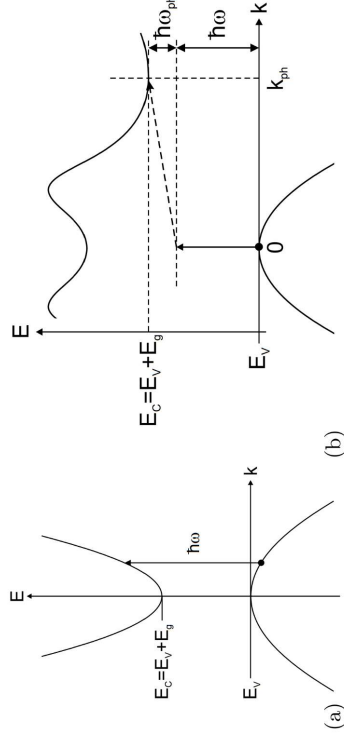
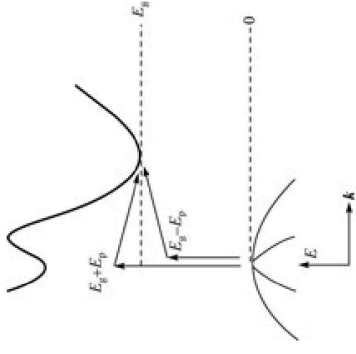


Fig. 9.6. (a) Direct optical transition and (b) indirect optical transitions between valence and conduction bands. The photon energy is $\hbar\omega$. The indirect transition involves a phonon with energy $\hbar\omega_{ph}$ and wavevector k_{ph} .

(b) Indirect transition

- The band extrema do not occur at the same wave vector k , the material is referred to as an indirect-gap semiconductor.
- For momentum conservation in such semiconductors, optical transitions (absorption and photoluminescence) are indirect, a participation of an extra particle, i.e., phonon, is required.
- The probability of optical transitions (absorption and luminescence) is substantially lower, compared with direct transitions. Therefore, fundamental absorption in indirect-gap semiconductors is usually weaker as compared with direct-gap materials.



Absorption with phonon emission

$$\hbar\omega_c = E_c(k_c) - E_v(k_v) + E_p \quad (13.2.11a)$$

$$\hbar\omega \leq E_g + E_p \text{ 时 } \alpha_c(\hbar\omega) = 0 \quad (13.2.11b)$$

Absorption with phonon absorption

$$\hbar\omega_a = E_c(k_c) - E_v(k_v) - E_p \quad (13.2.12a)$$

$$\hbar\omega \leq E_g - E_p \text{ 时 } \alpha_a(\hbar\omega) = 0 \quad (13.2.12b)$$



Absorption coefficient with phonon emission

$$\alpha_c(\hbar\omega) = \begin{cases} \frac{B(\hbar\omega - E_g - E_p)^2}{1 - \exp(-E_p/k_B T)}, & \hbar\omega > (E_g + E_p) \\ 0, & \hbar\omega \leq (E_g + E_p) \end{cases} \quad (13.2.13)$$

Absorption coefficient with phonon absorption

$$\alpha_a(\hbar\omega) = \begin{cases} \frac{B(\hbar\omega - E_g + E_p)^2}{\exp(E_p/k_B T) - 1}, & \hbar\omega > (E_g - E_p) \\ 0, & \hbar\omega \leq (E_g - E_p) \end{cases} \quad (13.2.14)$$

The total absorption coefficient with phonon emission and absorption

$$\alpha_i(\hbar\omega) = \begin{cases} B \left[\frac{(\hbar\omega - E_g - E_p)^2}{1 - \exp(-E_p/k_B T)} + \frac{(\hbar\omega - E_g + E_p)^2}{\exp(E_p/k_B T) - 1} \right], & \hbar\omega > (E_g + E_p) \\ B \frac{(\hbar\omega - E_g + E_p)^2}{\exp(E_p/k_B T) - 1}, & E_g - E_p \leq \hbar\omega \leq E_g + E_p \\ 0, & \hbar\omega \leq (E_g - E_p) \end{cases} \quad (13.2.15)$$



Absorption coefficient

$$\alpha(\hbar\omega) = A \sum_{if} W_{if}^{ab} n_i n_f' \quad (13.2.8)$$

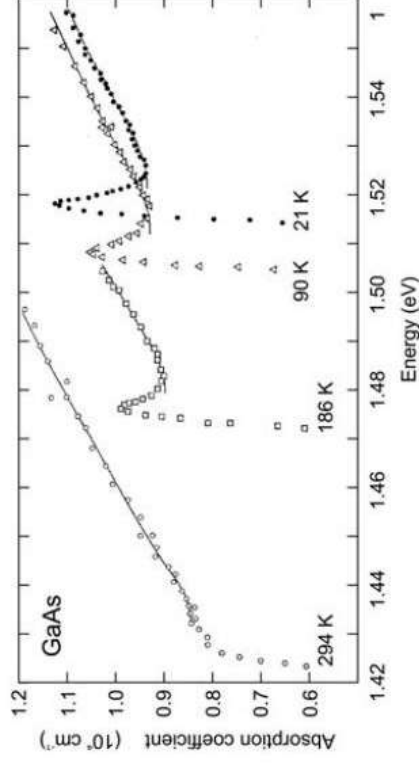
$$\alpha_d(\hbar\omega) = \begin{cases} A(\hbar\omega - E_g)^{1/2}, & \hbar\omega \geq E_g \\ 0, & \hbar\omega < E_g \end{cases} \quad (13.2.9a)$$

$$A \approx \frac{(2m_r^*)^{3/2} e^2}{\eta c m_e^* \hbar \epsilon_0} \quad (13.2.9b)$$

$$m_r^* = \frac{m_h^* m_e^*}{m_h^* + m_e^*} \quad (13.2.9c)$$

$$m_e^* = m_0, \eta = 4, A \approx 2 \times 10^4 \text{ cm}^{-1}$$

$$\alpha_d(\hbar\omega) = \begin{cases} \approx 2 \times 10^4 (\hbar\omega - E_g)^{1/2}, & \hbar\omega \geq E_g \\ 0, & \hbar\omega < E_g \end{cases} \quad (13.2.10)$$



Absorption spectra of GaAs close to the bandgap at different temperatures



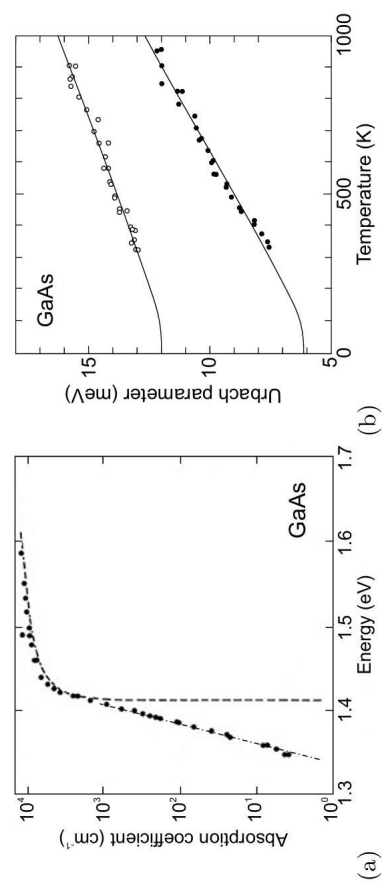
Urbach Tail

- Instead of the ideal $(E - E_g)^{1/2}$ dependence of the direct band-edge absorption, often an exponential tail is observed. This tail is called the Urbach tail, and follows the functional dependence

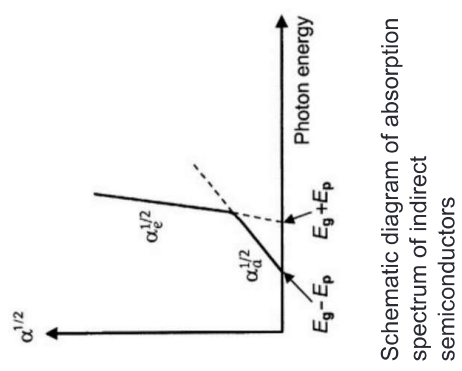
$$\alpha(E) = \alpha_g \exp\left(\frac{E - E_g}{E_0}\right),$$

where E_0 is the characteristic width of the absorption edge, the so-called Urbach parameter

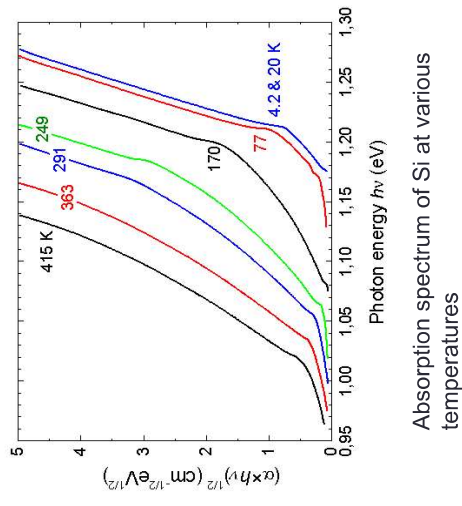
- The Urbach tail is attributed to transitions between band tails below the band edges. Such tails can originate from disorder of the perfect crystal, e.g. from defects or doping, and the fluctuation of electronic energy bands due to lattice vibrations.



Experimental absorption spectrum (circles) of GaAs at room temperature on a semilogarithmic plot. The exponential tail below the bandgap is called the Urbach tail (the dash-dotted line corresponds to $E_0 = 10.3$ meV). (b) Temperature dependence of Urbach parameter E_0 for two GaAs samples. Experimental data for undoped (solid circles) and Si-doped ($n = 2 \times 10^{18} \text{ cm}^{-3}$, empty circles) GaAs and theoretical fits (solid lines) with one-phonon model.

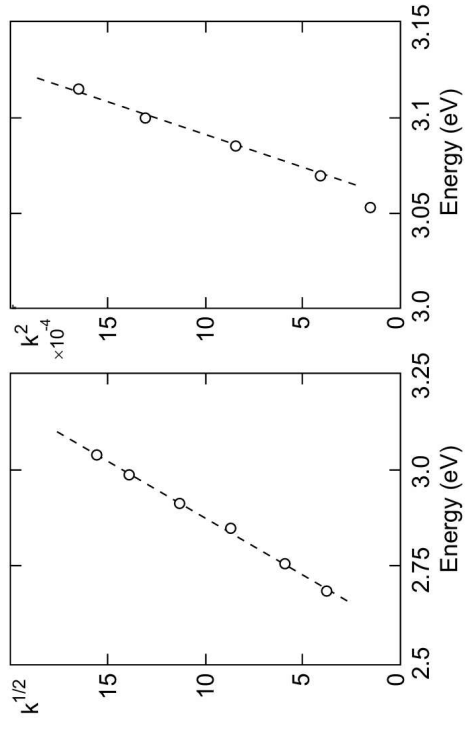


Schematic diagram of absorption spectrum of indirect semiconductors



Absorption spectrum of Si at various temperatures

Direct absorption in indirect semiconductors



Absorption of BaTiO₃ at room temperature. Experimental data (circles) with fits (dashed lines) $\propto E^2$ and $\propto E^{1/2}$, respectively.

■ 13.3.2 Excitons in Organic Semiconductors and in Insulators

- Roughly 600 inorganic semiconductors but only a dozen or so organic ones .
 - Hoped that applications as cheap, large area luminescence devices and as electronic devices will be developed.
 - Crystalline organic semiconductors usually contain several molecules per unit cell, e.g., anthracene.
 - The spectrum of lattice vibrations is complex, with two types of modes: acoustic and optic modes and vibrons or intramolecular modes.
 - Concerning the electrons, there are valence bands and conduction bands with a gap between them.
 - The bands are rather narrow, resulting in large effective masses of the charge carriers and low mobilities ($1\sim 10\text{ cm}^2/\text{Vs}$)
 - The interband excitations form excitons with some characteristic difference from the (Wannier) excitons in most inorganic semiconductors.
 - Deals in the wide gap ones with Frenkel excitons, electrons and holes reside in the same molecule, resulting in single-triplet splittings.
- The triplet states: low oscillator strength, low luminescence yield and long lifetime(ms).
- The single states: high oscillator strength, and short lifetime(ns).
- Coupling of excitons to phonons may be strong, possibly leading to self-trapping.



13.4 Recombination process

13.4.1 Introduction

- In semiconductors, various excitations (e.g., photon or electron irradiation) may lead to the generation of charge carriers in excess of the thermal equilibrium densities.
- Recombination of electron-hole pairs restores that equilibrium. Recombination processes are distinguished as **radiative** or **nonradiative**, depending on whether the recombination results in the emission of a photon or not, respectively
- Luminescence is generally described in terms of radiative recombination of electron-hole pairs, which may involve transitions between states in the conduction or valence bands and those in the energy gap due to donors and acceptors
- Depending on the source of excitation, the luminescence process can be categorized as **photoluminescence** (photon excitation), **electroluminescence** (excitation by application of an electric field), and **cathodoluminescence** (excitation by cathode rays, or energetic electrons).
- In **direct-gap semiconductors** (e.g., GaAs and CdS), the most likely (direct) transitions are across the minimum energy gap, between the most probably filled states at the minimum of the conduction band and the states most likely to be unoccupied at the maximum of the valence band. Radiative recombination between electrons and holes is relatively likely in such transitions
- In **indirect-gap semiconductors** (e.g., Si and GaP), since the recombination of electron-hole pairs requires a participation of an extra particle (i.e., phonon), the probability of such a process is significantly lower as compared with direct transitions. Thus, fundamental emission in indirect-gap semiconductors is relatively weak, especially when compared with that due to impurities or defects.



13.3 Exciton enhancement of optical absorption

13.3.1 Exciton-Photon Coupling

- Weak coupling: irreversible absorption or recombination
- Strong coupling-Exciton polariton

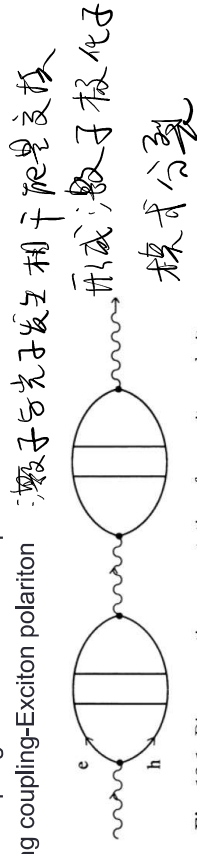


Fig. 13.1. Diagrammatic representation of an exciton polariton



John J. Hopfield, born in Chicago, Illinois, USA in 1933. He received his PhD in Physics from Cornell University in 1958 and currently serves as a professor at Princeton University



和激子相关的吸收

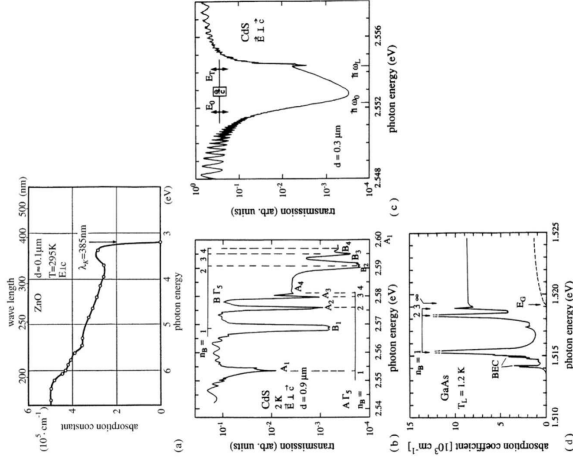


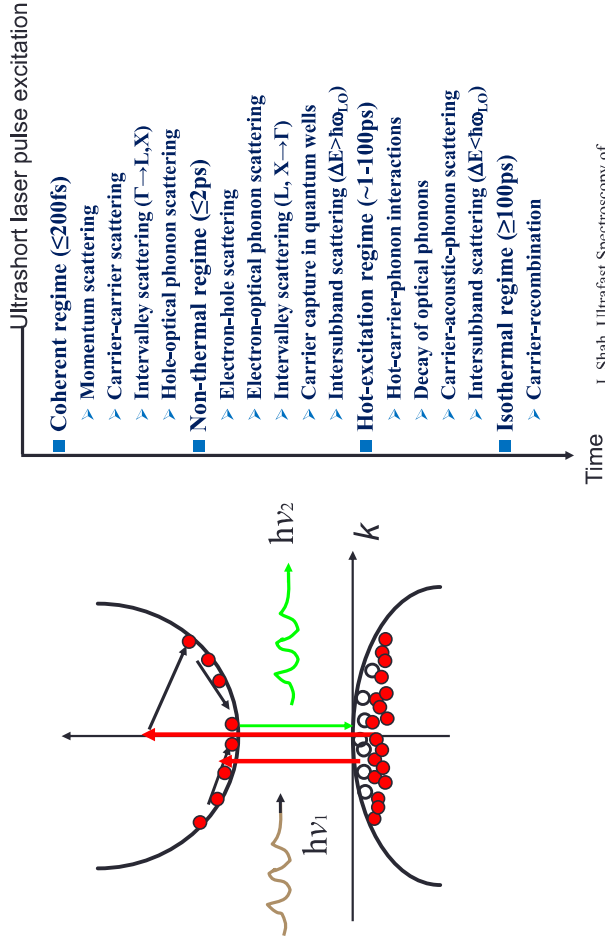
Fig. 13.9. An experimental absorption spectrum of ZnO at RT (a), an experimental transmission spectrum of CdS for the orientation $E_{\perp}$ in the region of the A and B exciton resonances (b); a calculated one for the $A\Gamma_5$ resonance for $m_0 = 1$ (c); and an absorption spectrum for a thin GaAs sample (d). According to [43M1] to [39V1, 82R1] and to [91U1], respectively

➢ The absorption spectra:

- 13.9a an absorption spectrum of ZnO layer at RT
- 13.9b experimental transmission spectra are shown A and $B\Gamma_5$ resonance of a thin CdS platelet type sample
- 13.9c shows a calculated transmission spectrum of the $A\Gamma_5$ $n_B = 1$ resonance
- 13.9d overview of the absorption spectrum of a thin GaAs sample.

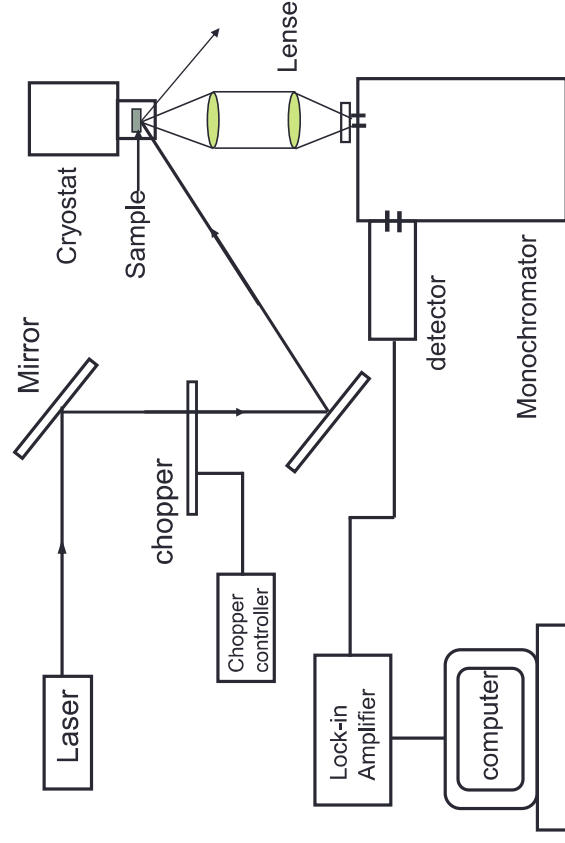


Carrier Relaxation: Four regimes



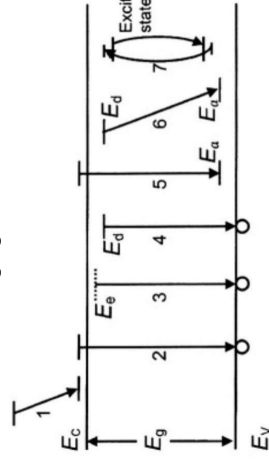
J. Shah, Ultrafast Spectroscopy of Semiconductors and Nanostructures (Springer, 1998)

13.4.3 Photoluminescence spectroscopy



13.4.2 Radiative Transitions

A simplified schematic diagram of radiative recombinations in semiconductors containing impurities is presented in the following figure.



Schematic diagram of radiative recombinations between the conduction band E_c , the valence band E_v , and exciton E_x , donor E_d , acceptor E_a levels in a semiconductor.

- In process 1**, which is intraband transition, an electron excited well above the conduction-band edge dribbles down and reaches thermal equilibrium with the lattice. This thermalization process may lead to phonon-assisted photon emission or, more likely, phonon emission only.
- In process 2**, which is interband transition, direct recombination between an electron in the conduction band and a hole in the valence band results in the emission of a photon of energy $\hbar\omega \approx E_g$; this transition produces intrinsic luminescence. Although this recombination occurs from states close to the corresponding band edges, the thermal distribution of carriers in these states will lead, in general, to a broad emission spectrum.

- BEC often show up in luminescence and absorption spectra as extremely narrow peaks.
- Chemical shift or central cell correction : a splitting of the lines caused by the chemical nature of the binding atom.
- the low temperature luminescence of high quality samples is generally dominated by defect (or localisation-) related recombination processes.

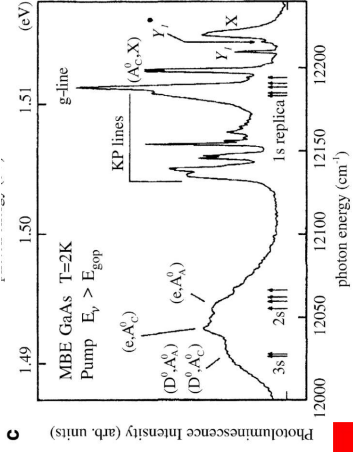


Fig. 14.2 Low-temperature and low-excitation luminescence spectra of ZnO (a) of two differently grown ZnSe samples (solid line: MBE growth with element sources, dashed line: with a compound source) (b) and of GaAs (c)

■ Recombination process : free-to-bound transition

The corresponding emission peak is at :

$$\hbar\omega_{FB} = E_g - E_{D^0/A^0}^b - m\hbar\omega_{LO}$$

■ Recombination process with DA pairs

The corresponding emission peak is at :

$$\hbar\omega_{FB} = E_g - E_{D^0}^b - E_{A^0}^b - m\hbar\omega_{LO}$$

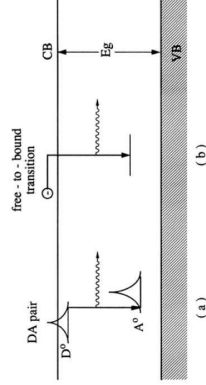


Fig. 14.9 A donor-acceptor pair (a) and a free-to-bound recombination process (b)

Fig. 14.10. Donor-acceptor pair luminescence in CdS including some phonon replicas [79B1]

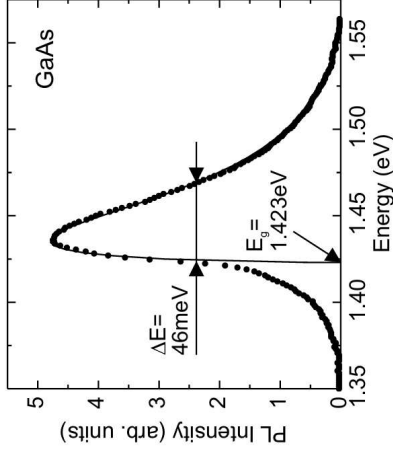
□ The spontaneous recombination rate

$$\begin{aligned} r_{sp}(E) &= \int_{E_C}^{\infty} dE_e \int_{-\infty}^{E_V} dE_h C(E_e, E_h) \\ &\times D_e(E_e) f_e(E_e) D_h(E_h) f_h(E_h) \delta(E - E_e + E_h) \\ &= \int_{E_C}^{E_V+E_V} dE_e C(E_e, E_e - E) \\ &\times D_e(E_e) f_e(E_e) D_h(E_e - E) f_h(E_e - E) \end{aligned}$$

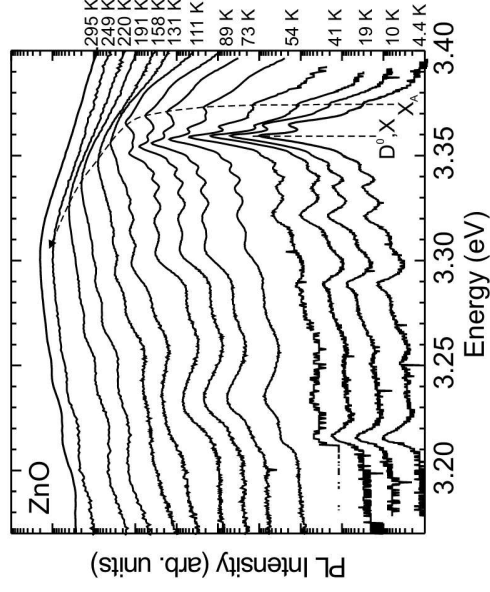
At small excitation and at low doping it can be approximated by the Boltzmann distribution function and the lineshape is given as

$$I(E) \propto (E - E_g)^{1/2} \exp\left(-\frac{E}{kT}\right) \quad (13.4.1)$$

Photoluminescence spectrum of an undoped LPE-grown epitaxial GaAs layer at room temperature and low cw ($\lambda = 647\text{nm}$) excitation density ($10\text{W}/\text{cm}^2$). The solid line is a lineshape fit with (13.4.1).



■ Free exciton recombination



Temperature-dependent luminescence spectra of a ZnO thin film (on sapphire). At low temperatures, the spectra are dominated by donor-bound exciton transitions. The vertical dashed line indicates the low-temperature position of the donor-bound exciton transition (D⁰X). The curved dashed line visualizes the energy position of the free-exciton transition (X_N) that becomes dominant at room temperature.

Further discussion on the SRH recombination rate equation

$$R_{\text{SRH}} = \frac{N_t \sigma_n \sigma_p v_{\text{th}} (np - n_i^2)}{\sigma_n \{n + n_i \exp[(E_t - E_i)/k_B T]\} + \sigma_p \{p + n_i \exp[(E_i - E_t)/k_B T]\}}$$

σ_n and σ_p are the electron and hole capture cross-sections, respectively

N_t is the volume density of deep levels

v_{th} is the carrier thermal velocity

E_t is the energy level of the trap

R has units of $\text{cm}^{-3}\text{s}^{-1}$

The term $np - n_i^2$ in this equation indicates the deviation from the thermal equilibrium conditions, and for $np = n_i^2$ (i.e., for the thermal equilibrium condition) the recombination rate is 0



13.4.4 Recombination rate

Various excitations may lead to the generation of charge carriers in excess of the thermal equilibrium densities, and the recombination of electron-hole pairs restores that equilibrium. The recombination rate R is proportional to the product of the concentration of occupied states (i.e., electrons) in the conduction band and the concentration of unoccupied states (i.e., holes) in the valence band. $R = B n_0 p_0$,

where B is a constant and n_0 and p_0 are the equilibrium concentrations of electrons and holes. If Δn and Δp are the excess carrier concentrations, we can write

$$-\frac{d\Delta n(t)}{dt} = B[n_0 + \Delta n(t)][p_0 + \Delta p(t)] - B n_i^2 \quad (13.4.5)$$

For a p-type semiconductor, this equation can be simplified to

$$-\frac{d\Delta n(t)}{dt} = B p_0 \Delta n(t) \quad (13.4.6)$$

Thus, if the generation process ceases, the initial excess carrier concentration will decay exponentially

$$\Delta n(t) = \Delta n(0) \exp(-B p_0 t) = \Delta n(0) \exp(-t/\tau_n) \quad (13.4.7)$$

Where $\tau_n = (B p_0)^{-1}$ is called the recombination lifetime or minority carrier lifetime. Similarly, the decay of excess holes in n-type semiconductor occurs with decay constant, $\tau_p = (B n_0)^{-1}$



Indirection transitions in indirect semiconductor

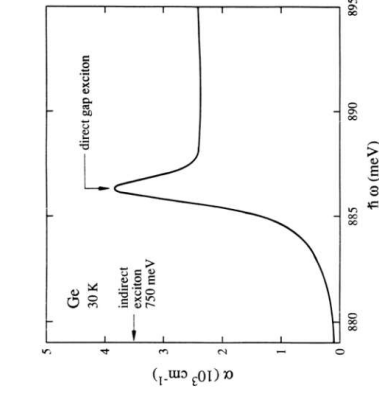


Fig. 13.34. Absorption around the direct exciton in Ge. According to [882]

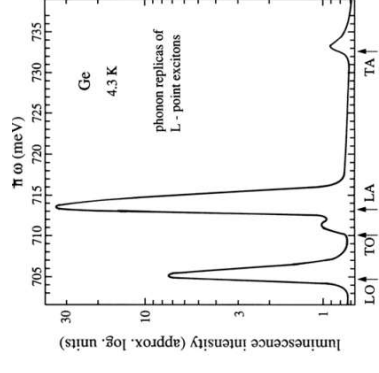


Fig. 13.35. A luminescence spectrum of Ge showing various phonon satellites. According to [7671]



13.4.3 Nonradiative recombinations

The electron-hole pair recombination may occur via nonradiative processes as well.

Nonradiative recombination processes include:

- multiple-phonon emission - direct conversion of the energy of an electron to heat
- Auger effect- the energy of an electron transition is absorbed by another electron which is raised to a higher-energy state in the conduction band, with subsequent emission of the electron from the material or dissipation of its energy through emission of phonons
- recombination due to surface states and defects

The kinetics of the recombination processes, related to the minority carrier capture at defects having levels in the energy gap of the semiconductor, can be described by the Shockley-Read-Hall (SRH) recombination model. According to this model, for a single trapping energy level present in the energy gap, the SRH recombination rate can be expressed as

$$R_{\text{SRH}} = \frac{N_t \sigma_n \sigma_p v_{\text{th}} (np - n_i^2)}{\sigma_n \{n + n_i \exp[(E_t - E_i)/k_B T]\} + \sigma_p \{p + n_i \exp[(E_i - E_t)/k_B T]\}} \quad (13.4.4)$$



- 因在贝尔实验室期间与同事共同发明晶体管，获得1956年诺贝尔物理学奖。晶体管在很大程度上取代了体积更大、效率更低的真空管，开创了微型电子时代，他还获得了90多项发明专利
- 1955年，他在加州创立了「肖克利实验室股份有限公司」，聘用了很多年轻优秀的人才。但很快肖克利个人的管理方法因其公司内部不合，八名主要员工（八叛逆）于1957年集体跳槽成立了仙童半导体公司，仙童半导体发明了集成电路



- In general, the description of such optical processes as the formation of the luminescence involves the analysis of the **generation**, **diffusion**, and **recombination of minority carriers**. Under equilibrium conditions, the **generation** of electron–hole pairs is balanced by **recombination** processes. These recombination processes may include radiative band-to-band and band-to-impurity recombination, and nonradiative recombination processes such as trap-assisted recombination, Auger recombination, and surface recombination.
- For both competitive **radiative** and **nonradiative** centers present, the observable lifetime is

$$\frac{1}{\tau} = \frac{1}{\tau_r} + \frac{1}{\tau_{nr}} \quad (13.4.8)$$
- In general, τ_{nr} is the resultant of several nonradiative recombination processes that may be present in a given material,

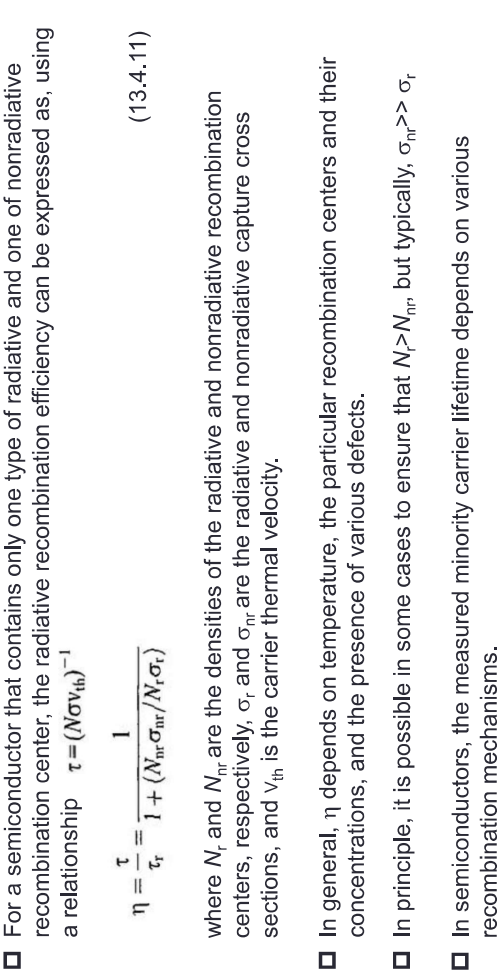
$$\frac{1}{\tau_{nr}} = \sum_i \frac{1}{\tau_{nri}} \quad (13.4.9)$$
- The radiative recombination efficiency (or internal quantum efficiency) η which is defined as the ratio of the radiative recombination rate R_r to the total recombination rate R , can be written as

$$\eta = \frac{\tau}{\tau_r} = \frac{1}{1 + (\tau_r/\tau_{nr})} \quad (13.4.10)$$
- When the nonradiative recombination process is faster, radiative recombination efficiency is relatively small. On the other hand, when the radiative recombination process is faster, radiative recombination efficiency is relatively greater

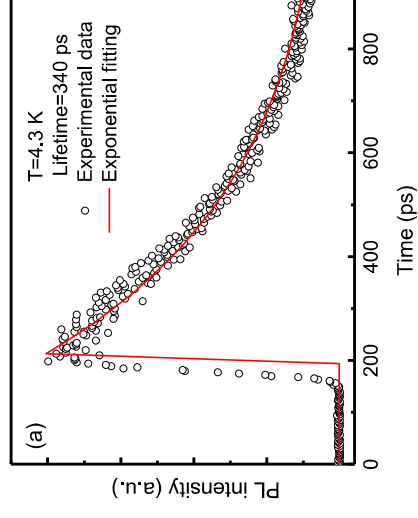
$$\frac{1}{\epsilon} = \frac{1}{\epsilon} + \frac{1}{\epsilon} \quad (13.4.12)$$

$$\frac{1}{x} = \frac{1}{x} + \frac{1}{x} + \frac{1}{x} \quad (13.4.13)$$

- ❑ In general, whereas τ_{bulk} depends on the density of recombination centers in the bulk of the material, τ_{surf} is determined to a large extent by the recombination centers on the surface.
- ❑ Surface states arise due to the abrupt change at the surface of the three-dimensional band structure associated with the bulk of the material.
- ❑ In addition, impurity atoms and oxide layers (on the surface of the material) may also produce discrete energy levels.
- ❑ In pure crystals, τ_{bulk} is typically long, so that surface recombination will dominate. In such a case, the lifetime of the material may depend strongly on surface treatment procedures.



PL decay of ZnO



$$I_{\text{PL}} = I_0 \exp(t/\tau)$$